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Task Description:	The space radiation environment poses a range of potential health risks to astronauts. Exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can damage critical cellular targets leading to a number of deleterious effects. Space travel has also been found to elicit changes in the redox state of cells that may exacerbate these adverse effects. These risks may be particularly problematic to stem/precursor cells within the CNS, where exposure to irradiation may promote genomic instability which could result in cancer and/or cognitive dysfunction. In vitro and in vivo we have shown neural precursor cells are exquisitely sensitive to irradiation, undergoing a dose-dependent induction of apoptosis that is associated with a chronic oxidative stress. These effects are strongly correlated with a persistent inhibition of neurogenesis that may underlie cognitive deficits observed after cranial irradiation. We have also shown that human brain tumors contain a significant number of cells that express many of the same immature lineage-specific markers common to neural precursor cells, supporting a possible role for these cells in the development of brain tumors. Based on these observations, we hypothesize that neural precursor cells exposed to high energy ions will be particularly susceptible to genomic instability and in a manner that depends upon redox state. Genomic instability is a process believed to drive cancer progression, and recently, redox state has been implicated in the induction and perpetuation of genomic instability. Redox state has been found to be a critical determinant in regulating the response of the CNS to a variety of damaging treatments, including irradiation. Changes in the redox environment are often associated with alterations in cellular metabolism which can promote oxidative stress, and these types of changes are common to tumors including those within the CNS. Because exposure to radiation has been found to cause genomic instability, and since oxidative stress is associated with the radiation response of neural precursor cells, it follows that changes in redox state might constitute a biochemical mechanism for modifying the radioresponse of the CNS. We are proposing a series of studies to test the overall hypothesis that alterations in cellular redox state promote genomic instability that can adversely impact the CNS upon exposure to high LET radiation. While the development of genomic instability may predispose neural precursor cells to transformation events in the CNS, the unstable phenotype may also lead to radioresistance that changes the phenotypic fate of neural precursor cells that impairs neurogenesis. We propose that changes in redox state play a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo; changes that will foster the onset of radiation-induced genomic instability, radioresistance and altered neurogenesis. Using an in vitro and in vivo approach the goals of our project will be addressed in the following 3 specific aims: Specific aim 1: Elucidate the dose-response relationship for the induction of genomic instability in neural precursor cells cultured in vitro or isolated from mouse brain after exposure to protons and iron ions. Specific aim 2: Determine if chronic changes in cellular redox state alter the dose response relationship determined above and predispose neural precursor cells to proton- and iron ion-induced genomic instability. Specific aim 3: Determine if differentiation of genomically unstable neural precursor cells is altered in vitro, or if neurogenesis is altered in vivo when unstable precursor cells are implanted into the brains of control or irradiated mice. Task Objective: To determine whether changes in redox state adversely impact the CNS. We will assess whether oxidative stress alters the radiosensitivity of neural precursors and predisposes them radiation-induced genomic instability. Experiments will ascertain whether such changes alter precursor cell plasticity in a manner that compromises their normal function. These studies will promote an understanding of the space radiation effects on the CNS. Data collected from these studies will improve both our understanding of the basic biology underlying CNS radiation effects, and in assessing risks to astronauts engaged in space travel. This work will establish the extent to which redox related changes derived from environmental stressors (oxidative injury) and/or damaging agents (ionizing radiation) are problematic to astronauts and whether specific countermeasures can be developed to minimize such risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will shed light on many of the basic biological processes that are associated with exposure of the CNS to ionizing radiation. As such, our work is directly relevant to those involved in the treatment of CNS tumors via radiotherapy. Such individuals include clinicians, basic scientists and even patients, all of whom have a vested interest in minimizing side-effects associated with the exposure of normal tissue to ionizing radiation. Outside of the clinical relevance, our work should foster a general interest to those members of the public supporting the long-term goals of NASA involving the expoloration and colonization of space.
	Summary of Accomplishments: We are now well into the funding period for this grant and we are pleased with much of our progress. Over this last year we have made new strides into areas of genomic instability that reflect the recent and rapid emergence of concepts regarding cancer stem cells. Genomic instability has been proposed as a mechanism capable of driving cells through the requisite steps along the path toward malignancy. As defined, genomic instability represents the increased rate of acquisition of genetic changes, and has come to encompass an impressive array of endpoints ranging from karyotypic abnormalities to changes in cloning efficiency. While certain questions remain regarding the causal role of genomic instability in cancer, there remains little doubt that once initiated, genomic instability can cause a wide range of alterations to cellular phenotypes induced though an assortment of dynamic changes to the genome. More recent studies have now suggested that possibly all cancers may be derived from a small subpopulation of cancer stem cells. Similar to normal stem cells, cancer stem cells can self renew, but instead of regenerating the normal cell types that define a specific tissue, cancer stem cells regenerate or “phenocopy” the aberrant phenotypes that characterize a particular neoplasm upon serial transplantation into recipient hosts. The idea that cancer is driven by it’s own self-renewing stem cell may or may not be consistent with the tenets of genomic instability, but resolving this issue has important mechanistic implications for understanding how cancer develops and evolves over time. For a cancer stem cell, genomic instability would present a potential threat to its capability to indefinitely self renew and recapitulate the representative phenotypes that characterize a particular neoplasm. Therefore, we propose that genomic instability will reduce the capability of cancer stem cells to retain their stem cell properties over time. Interestingly, this scenario predicts that a cancer stem cell undergoing genomic instability will no longer function as a stem cell, and under this circumstance, genomic instability may actually provide a mechanism for terminating rather than promoting cancer progression. To circumvent such potential problems, we propose that cancer stem cells are more resistant to radiation-induced damage/changes that promote cell kill and genomic instability, and that the various

Task Progress:	manifestations of genomic instability will be largely restricted to the pool of more radiosensitive progenitor cells. Thus, the underlying hypothesis of this grant will be to test whether cancer and normal stem cells derived from the brain are radioresistant and less likely to become genomically unstable compared to their progenitor cell progeny. We have had a great deal of success incorporating these new concepts into the aims of our NASA grant, and based on our new data, the scope of our overarching hypothesis becomes even more compelling, as we try to delineate the interplay between genomic instability and oxidative stress between various types of multipotent neural cells in the CNS. Many of these new findings are now presented at the end of this section and highlight many of the similarities between neural precursor cells and cancer stem cells, suggesting that the resident pool of multipotent cells in the CNS may represent latent reservoirs of potentially problematic cancer stem cells. Much of our now published work has clearly substantiated a major hypothesis of our overall grant, namely, that oxidative stress has a significant impact on changing cellular physiology and the radioresponse of multipotent neural stem and precursor cells. Our data demonstrates that altered redox state impacts cellular proliferation, mitochondrial function and radiosensitivity, parameters critical in impacting both normal tissue responses to irradiation but also carcinogenic processes that might predispose multipotent neural cells to transformation. We have leveraged our capability to alter the redox status of neural precursor cells endogenously (via cell density changes), metabolically, (via inhibition of electron transport), and environmentally (via addition of exogenous oxidants), to analyze how changes in oxidative stress impacts the function of neural precursor before and after irradiation. These methodologies have revealed the dual nature of oxidative stress. While acute increases in ROS are initially mitogenic, persistent increases in oxidative stress are growth inhibitory and radiosensitizing. Our implementation of exogenous oxidants such as hydrogen peroxide or mitochondrial electron transport chain inhibitors (antimycinA, menadione) that elevate mitochondrial-derived ROS also reveal that under conditions of prolonged oxidative stress, cells become more sensitive to the action of ionizing radiation. Work conducted at Loma Linda University has also demonstrated qualitatively similar results where conditions used to increase oxidative stress were found to enhance the sensitivity of neural precursors to 250 MeV protons. Collectively, these positive results suggest that interventions targeted to disrupt the metabolic state of cells and redox sensitive pathways are likely to impact genomic instability, differentiation and neurogenesis in the CNS. As alluded to above, some very recent data points to some intriguing similarities between primary neural precursor cells and brain tumor stem cells. A comparison between two glioma lines and neural precursor cells has revealed that these cells have similar post-irradiation redox responses. Both gliomas and neural precursor cells exhibited similar increases in oxidative stress following irradiation, a trend that was distinctly opposite when normal human brain cells were analyzed. Furthermore, neural precursor cells were found to exhibit similar phenotypic expression profiles as glioma cells. Recent fractionation studies conducted during our last BNL run (fall 07) now show that the radioresistant fraction of neural precursor cells can be enriched after heavy ion (56Fe) irradiation that express the CD133 cancer stem cell marker. Interestingly, survival and sphere forming assays performed on heavy ion irradiated cultures reveals two differentially sensitive subpopulations: a stem cell fraction that is relatively radioresistant and a precursor cell fraction that is more radiosensitive. The foregoing findings point to a number of shared properties between neural precursor cells and brain tumor stem cells, and suggests again that neural precursor cells may be latent cancer stem cells in the brain. Major accomplishments are summarized below: 1) In vitro studies using hippocampal precursors have revealed that endogenous oxidative stress increases the sensitivity of these cells to X-rays. 2) In vitro studies using hippocampal precursors have revealed that exogenous oxidative stress increases the sensitivity of these cells to X-rays and to 250 MeV protons. 3) In vitro studies have shown that perturbation of mitochondrial electron transport increases oxidative stress and enhances the sensitivity of neural precursors to irradiation. 4) Comparisons between neural precursor cells and brain tumor cells have shown that these cells share similar redox properties after irradiation. Neural precursor cells and gliomas show marked increases in oxidative stress after irradiation, an effect not found when normal brain cells were analyzed after irradiation. 5) Neural precursor cells and brain tumor cells express a number of shared phenotypic markers. The similar expression of immature markers suggest that a fraction of brain tumors contain cells similar to neural precursor cells. 6) Normal neural stem and precursor cells exhibit CD133 expression, a marker reported to be specific for brain tumor stem cells. 7) Dose fractionation of normal subventricular stem and precursor cells using either x-rays or heavy ions selects for a radioresistant subpopulation enriched for CD133 positive cells. 8) Cell survival and neurosphere assays reveals neural stem cells to be more radioresistant than their precursor cell progeny. 9) We have started of our breeding colony of Tel+/- mice at UCI. 10) Mice heterozygous and homozygous for the tagged chromosome 11 will be bred for in vivo studies (BNL spring 08) and for the generation of neural stem and precursor primary cell cultures. 11) We have published 4 manuscripts, with three more nearing completion by mid-08. 12) NASA funding has led 18 formal teaching and research seminars presented internationally and nationally.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
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